

# Advanced Optimization

## Chapter 5 — Conjugate Gradient Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 5

# Two reasons to care, and where CG lives in this course

## As a linear solver (§5.1)

Hestenes–Stiefel, 1950s. Solves  $Ax = b$ ,  $A \succ 0$ , without factorization. Performance governed by the *eigenvalue distribution* of  $A$  — which *preconditioning* can reshape.

## As an optimizer (§5.2)

Fletcher–Reeves, 1960s. One of the earliest large-scale nonlinear methods. **No matrix storage**, only a few vectors; faster than steepest descent.

## Where the linear CG reappears later

- ▶ Ch. 4 / Ch. 7: CG–Steihaug for the trust-region subproblem, Newton-CG
- ▶ Ch. 16: projected CG for equality-constrained QP
- ▶ Ch. 19: iterative solution of the interior-point KKT system at large scale

Every one of these needs the properties proved in §5.1, and the preconditioning idea at the end of it.

# One problem, two lenses

$$Ax = b \quad (5.1) \quad \Longleftrightarrow \quad \min_x \phi(x) = \frac{1}{2}x^T Ax - b^T x \quad (5.2),$$

$A$  symmetric positive definite. Same unique solution  $x^*$ .

The gradient of  $\phi$  is the **residual** of the linear system:

$$\nabla \phi(x) = Ax - b =: r(x), \quad r_k = Ax_k - b. \quad (5.3-5.4)$$

## Consequence for reading this chapter

Every statement about “residuals” is a statement about gradients, and every “minimize  $\phi$  along  $p$ ” is an exact line search on a quadratic. We switch lenses freely.

Exact minimizer of  $\phi$  along  $x_k + \alpha p_k$  (from Ch. 3, (3.55)):

$$\alpha_k = -\frac{r_k^T p_k}{p_k^T A p_k}.$$

# Conjugate directions

## Definition

Nonzero vectors  $\{p_0, \dots, p_l\}$  are **conjugate** with respect to  $A \succ 0$  if

$$p_i^T A p_j = 0 \quad \text{for all } i \neq j. \quad (5.5)$$

Conjugacy is orthogonality in the inner product  $\langle u, v \rangle_A = u^T A v$ . A conjugate set is linearly independent (homework).

## Conjugate direction method

Given  $x_0$  and a conjugate set  $\{p_0, \dots, p_{n-1}\}$ :

$$x_{k+1} = x_k + \alpha_k p_k, \quad \alpha_k = -\frac{r_k^T p_k}{p_k^T A p_k}. \quad (5.6-5.7)$$

**Theorem 5.1:** for any  $x_0$ , this method reaches  $x^*$  in at most  $n$  steps. The proof is five lines, and it is the reason conjugacy matters — next slide.

# Why $n$ steps suffice (proof of Theorem 5.1)

The  $p_i$  span  $\mathbb{R}^n$ , so write

$$x^* - x_0 = \sigma_0 p_0 + \sigma_1 p_1 + \cdots + \sigma_{n-1} p_{n-1}.$$

Multiply by  $p_k^T A$ ; conjugacy kills every term but one:

$$\sigma_k = \frac{p_k^T A(x^* - x_0)}{p_k^T A p_k}. \quad (5.8)$$

Now the iterates:  $x_k = x_0 + \alpha_0 p_0 + \cdots + \alpha_{k-1} p_{k-1}$ , so again by conjugacy  $p_k^T A(x_k - x_0) = 0$ , hence

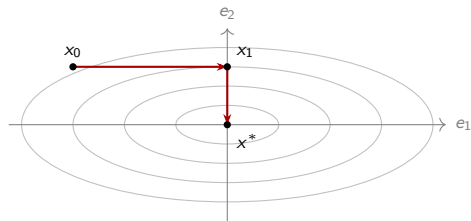
$$p_k^T A(x^* - x_0) = p_k^T A(x^* - x_k) = p_k^T (b - Ax_k) = -p_k^T r_k.$$

Compare with (5.7):  $\sigma_k = \alpha_k$ .  $\square$

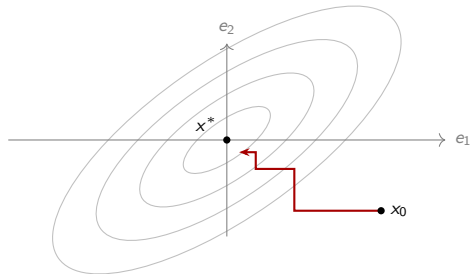
## What the proof says

Conjugacy *decouples* the coordinates of the error in the basis  $\{p_i\}$ . A 1-D minimization along  $p_k$  fixes the  $k$ -th coordinate exactly and never disturbs the others. In the right basis, an  $n$ -dimensional problem is  $n$  independent 1-D problems.

# The geometry (Figures 5.1, 5.2)



A diagonal: coordinate search finishes in  $n$  steps.



General A: coordinate search never finishes.

**Fix:** change variables  $\hat{x} = S^{-1}x$  with  $S = [p_0 \cdots p_{n-1}]$ . Then  $\hat{\phi}(\hat{x}) = \frac{1}{2}\hat{x}^T(S^TAS)\hat{x} - (S^Tb)^T\hat{x}$  and  $S^TAS$  is diagonal by conjugacy. Coordinate search in  $\hat{x}$ -space is the conjugate direction method in  $x$ -space.

# Expanding subspace minimization (Theorem 5.2)

From (5.4) and (5.6), the residual updates by

$$r_{k+1} = r_k + \alpha_k A p_k. \quad (5.10)$$

## Theorem 5.2

For the conjugate direction method from any  $x_0$ :

$$r_k^T p_i = 0, \quad i = 0, \dots, k-1, \quad (5.11)$$

and  $x_k$  minimizes  $\phi$  over the affine set  $x_0 + \text{span}\{p_0, \dots, p_{k-1}\}$  (5.12).

**Why.** A point minimizes  $\phi$  over (5.12) iff its residual is orthogonal to every  $p_i$  in the span (differentiate  $h(\sigma) = \phi(x_0 + \sum \sigma_i p_i)$ ). Induction:  $p_{k-1}^T r_k = p_{k-1}^T r_{k-1} + \alpha_{k-1} p_{k-1}^T A p_{k-1} = 0$  by the choice of  $\alpha_{k-1}$ ; for  $i < k-1$ ,  $p_i^T r_k = p_i^T r_{k-1} + \alpha_{k-1} p_i^T A p_{k-1} = 0$  by hypothesis and conjugacy.

**This is stronger than Theorem 5.1:** not just “done after  $n$  steps”, but “after  $k$  steps, optimal on a  $k$ -dimensional subspace”. Progress is monotone in a precise sense — and this is what makes CG useful when stopped early.

Any conjugate set works (eigenvectors, conjugate Gram–Schmidt) — but those cost  $O(n^2)$  storage or an eigendecomposition. The conjugate *gradient* method generates the set on the fly.

# The special property of conjugate gradients

Choose each direction as a combination of the current steepest descent direction and the *previous direction only*:

$$p_k = -r_k + \beta_k p_{k-1}, \quad (5.13)$$

with  $\beta_k$  fixed by requiring  $p_{k-1}^T A p_k = 0$ :

$$\beta_k = \frac{r_k^T A p_{k-1}}{p_{k-1}^T A p_{k-1}}.$$

Start with  $p_0 = -r_0$ .

## Algorithm 5.1 (CG, preliminary version)

```
 $r_0 \leftarrow Ax_0 - b, p_0 \leftarrow -r_0, k \leftarrow 0;$  while  $r_k \neq 0$ :  
   $\alpha_k \leftarrow -r_k^T p_k / p_k^T A p_k;$   $x_{k+1} \leftarrow x_k + \alpha_k p_k;$   $r_{k+1} \leftarrow Ax_{k+1} - b;$   
   $\beta_{k+1} \leftarrow r_{k+1}^T A p_k / p_k^T A p_k;$   $p_{k+1} \leftarrow -r_{k+1} + \beta_{k+1} p_k;$   $k \leftarrow k + 1.$ 
```

## The remarkable part

We enforced conjugacy with  $p_{k-1}$  only. Yet  $p_k$  turns out to be conjugate to *all* of  $p_0, \dots, p_{k-2}$  as well — automatically. That is why the method needs almost no storage. It is not obvious; it is Theorem 5.3.

# The four properties (Theorem 5.3)

Define the **Krylov subspace**  $\mathcal{K}(r_0; k) = \text{span}\{r_0, Ar_0, \dots, A^k r_0\}$  (5.15).

## Theorem 5.3

If  $x_k \neq x^*$ , then

$$r_k^T r_i = 0, \quad i < k, \quad (5.16)$$

$$\text{span}\{r_0, \dots, r_k\} = \mathcal{K}(r_0; k), \quad (5.17)$$

$$\text{span}\{p_0, \dots, p_k\} = \mathcal{K}(r_0; k), \quad (5.18)$$

$$p_k^T A p_i = 0, \quad i < k. \quad (5.19)$$

Hence  $x_k \rightarrow x^*$  in at most  $n$  steps.

- ▶ Residuals are mutually *orthogonal*; directions are mutually *conjugate*. “Conjugate gradient” is a misnomer — it is the search directions that are conjugate.
- ▶ Everything lives in the Krylov space: CG is a **Krylov subspace method**, and  $x_{k+1} = x_0 + P_k(A)r_0$  for some polynomial  $P_k$  of degree  $k$ . This view gives the rate.
- ▶ The proof (induction on  $k$ , using (5.10), (5.11)) **requires**  $p_0 = -r_0$ . Start anywhere else and the properties fail. Remember this when restarts come up in §5.2.

# The practical form (Algorithm 5.2)

Use (5.11) and (5.14e):  $r_k^T p_k = -r_k^T r_k$ , so  $\alpha_k = r_k^T r_k / p_k^T A p_k$ . Use (5.10) and (5.16):  $\alpha_k r_{k+1}^T A p_k = r_{k+1}^T (r_{k+1} - r_k) = r_{k+1}^T r_{k+1}$ , so  $\beta_{k+1} = r_{k+1}^T r_{k+1} / r_k^T r_k$ .

## Algorithm 5.2 (CG)

Given  $x_0$ ;  $r_0 \leftarrow Ax_0 - b$ ,  $p_0 \leftarrow -r_0$ ,  $k \leftarrow 0$ ;

**while**  $r_k \neq 0$ :

$$\begin{array}{lll} \alpha_k \leftarrow r_k^T r_k / p_k^T A p_k; & x_{k+1} \leftarrow x_k + \alpha_k p_k; & r_{k+1} \leftarrow r_k + \alpha_k A p_k; \\ \beta_{k+1} \leftarrow r_{k+1}^T r_{k+1} / r_k^T r_k; & p_{k+1} \leftarrow -r_{k+1} + \beta_{k+1} p_k; & k \leftarrow k + 1. \end{array}$$

(5.24)

- ▶ Per iteration: **one product**  $A p_k$ , two inner products, three vector updates. Storage:  $x$ ,  $r$ ,  $p$ ,  $A p$  — four vectors.
- ▶  $A$  is never modified, never factored, never filled in. Only  $p \mapsto A p$  is needed —  $A$  can be a function.
- ▶ For small  $n$ : use a factorization instead — less sensitive to rounding. CG is for *large* problems.

# Rate of convergence: the polynomial view

By (5.18),  $x_{k+1} = x_0 + P_k^*(A)r_0$  for a polynomial  $P_k^*$  of degree  $k$  (5.26). Measure error in the  $A$ -norm,  $\|z\|_A^2 = z^T A z$ , for which  $\frac{1}{2}\|x - x^*\|_A^2 = \phi(x) - \phi(x^*)$  (5.28) — the same norm as in Ch. 3.

Theorem 5.2 says  $x_{k+1}$  minimizes  $\phi$  over  $x_0 + \mathcal{K}(r_0; k)$ , i.e.

$$\|x_{k+1} - x^*\|_A = \min_{P_k} \|x_0 + P_k(A)r_0 - x^*\|_A. \quad (5.29)$$

CG is optimal among all methods confined to the Krylov space.

With  $r_0 = A(x_0 - x^*)$  and  $x_0 - x^* = \sum \xi_i v_i$  in the eigenbasis of  $A$ :

$$\|x_{k+1} - x^*\|_A^2 = \min_{P_k} \sum_i \lambda_i [1 + \lambda_i P_k(\lambda_i)]^2 \xi_i^2 \leq \min_{P_k} \max_i [1 + \lambda_i P_k(\lambda_i)]^2 \|x_0 - x^*\|_A^2. \quad (5.33)$$

The whole rate theory in one sentence

CG converges as fast as the best degree- $(k+1)$  polynomial with  $Q(0) = 1$  can be made small *on the eigenvalues of  $A$* . Rate depends on where the eigenvalues sit, not on  $n$ .

# Distinct eigenvalues and clusters (Theorems 5.4, 5.5)

## Theorem 5.4

If  $A$  has only  $r$  distinct eigenvalues, CG terminates in at most  $r$  iterations.

Proof:  $Q_r(\lambda) = \prod_j (1 - \lambda/\tau_j)$  vanishes on every eigenvalue and has  $Q_r(0) = 1$ ; take  $P_{r-1} = (Q_r - 1)/\lambda$  in (5.33).

## Theorem 5.5

With eigenvalues  $\lambda_1 \leq \dots \leq \lambda_n$ ,

$$\|x_{k+1} - x^*\|_A^2 \leq \left( \frac{\lambda_{n-k} - \lambda_1}{\lambda_{n-k} + \lambda_1} \right)^2 \|x_0 - x^*\|_A^2. \quad (5.35)$$

Idea: choose  $Q_{k+1}$  with roots at the  $k$  largest eigenvalues and one in the middle of the rest.



$m$  large eigenvalues, the rest within  $\epsilon$  of 1: after  $m + 1$  steps the error is  $\approx \epsilon \|x_0 - x^*\|_A$ . Roughly:  $r$  clusters  $\Rightarrow$  about  $r$  iterations. (Figures 5.4–5.5: sharp drops at the predicted iteration, sometimes one earlier — the bound is an upper bound.)

# The condition-number bound, and the comparison with steepest descent

With only the extreme eigenvalues known,  $\kappa(A) = \lambda_n/\lambda_1$ :

$$\|x_k - x^*\|_A \leq 2 \left( \frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1} \right)^k \|x_0 - x^*\|_A. \quad (5.36)$$

Compare steepest descent (3.29): same form, but with  $\kappa$  in place of  $\sqrt{\kappa}$ .

| $\kappa$ | SD factor $\left(\frac{\kappa-1}{\kappa+1}\right)$ | CG factor $\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)$ | iterations to reduce error $10^6 \times$ (CG / SD) |
|----------|----------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------|
| $10^2$   | 0.980                                              | 0.818                                                            | $\approx 70$ / $\approx 700$                       |
| $10^4$   | 0.9998                                             | 0.980                                                            | $\approx 700$ / $\approx 70,000$                   |
| $10^6$   | 0.999998                                           | 0.998                                                            | $\approx 7,000$ / $\approx 7,000,000$              |

- ▶ (5.36) is often a large *overestimate* — clustering (Thm 5.5) is what actually happens.
- ▶ But the  $\sqrt{\kappa}$  is real: CG is the SD method with a memory, and the memory buys a square root.
- ▶ Ill-conditioned problems still hurt. The lever we have is the eigenvalue distribution itself.

# Preconditioning

Change variables  $\hat{x} = Cx$  (5.37),  $C$  nonsingular. Then

$$\hat{\phi}(\hat{x}) = \frac{1}{2}\hat{x}^T(C^{-T}AC^{-1})\hat{x} - (C^{-T}b)^T\hat{x}, \quad (5.38)$$

and CG on  $\hat{\phi}$  converges according to the eigenvalues of  $C^{-T}AC^{-1}$ , not  $A$ .

## Goal

Choose  $C$  so that  $C^{-T}AC^{-1}$  has a *small condition number*, or better, *clustered eigenvalues*.  
Ideal:  $C^{-T}AC^{-1} \approx I$ .

The transformation is never carried out explicitly. Run CG in  $\hat{x}$ , translate every line back to  $x$ , and what remains is a method that uses only

$$M = C^T C \succ 0,$$

and only through solves  $My = r$ .

**Trade-off:**  $M$  should be close enough to  $A$  to help, cheap enough to form, and cheap enough to solve with. No single choice wins everywhere.

## Algorithm 5.3 (Preconditioned CG)

Given  $x_0$ , preconditioner  $M$ ;  $r_0 \leftarrow Ax_0 - b$ ; solve  $My_0 = r_0$ ;  $p_0 \leftarrow -y_0$ ;  $k \leftarrow 0$ ;

**while**  $r_k \neq 0$ :

$$\begin{aligned} \alpha_k &\leftarrow r_k^T y_k / p_k^T A p_k; & x_{k+1} &\leftarrow x_k + \alpha_k p_k; & r_{k+1} &\leftarrow r_k + \alpha_k A p_k; \\ \text{solve } My_{k+1} &= r_{k+1}; & \beta_{k+1} &\leftarrow r_{k+1}^T y_{k+1} / r_k^T y_k; & p_{k+1} &\leftarrow -y_{k+1} + \beta_{k+1} p_k. \end{aligned}$$

(5.39)

- ▶  $M = I$ : Algorithm 5.2 exactly.
- ▶ Extra cost per iteration: **one solve with  $M$** . That solve must be cheap — it is the whole design constraint.
- ▶ Orthogonality of residuals becomes  $r_i^T M^{-1} r_j = 0$ ,  $i \neq j$  (5.40).

Note that  $M$  appears only via  $y = M^{-1}r$ . So  $M$  can be given implicitly — as a factorization, as a coarse-grid solve, as a few sweeps of another iteration.

# Practical preconditioners

- ▶ **Problem-derived.** For PDE discretizations,  $My = r$  can be a *coarser* discretization of the same problem. Knowledge of where  $A$  came from is the best preconditioner design tool there is.
- ▶ **General-purpose.** SSOR, banded, and **incomplete Cholesky**: run Cholesky but keep the factor  $\tilde{L}$  as sparse as (roughly) the lower triangle of  $A$ . Then  $A \approx \tilde{L}\tilde{L}^T$ , set  $C = \tilde{L}^T$ , and

$$C^{-T}AC^{-1} = \tilde{L}^{-1}A\tilde{L}^{-T} \approx I.$$

Solving  $My = r$  is two sparse triangular solves — about the cost of one  $Ap$ .

- ▶ **Pitfalls of incomplete Cholesky.** The sparsified factor may fail to be positive definite (shift the diagonal); the factorization may break down (allow more fill — but then it is more expensive to form and apply).

## For the problems in this room

Power-flow Jacobians and OPF KKT systems are sparse and structured (bus admittance, reduced network). Whenever a solver reports “iterative linear solver”, the preconditioner is where the performance is decided — and where problem structure pays off.

## Fletcher–Reeves: two changes to Algorithm 5.2

- 1 Replace the exact step length by a *line search* on the nonlinear  $f$  along  $p_k$ .
- 2 Replace the residual  $r_k = \nabla\phi(x_k)$  by  $\nabla f_k$ .

### Algorithm 5.4 (FR)

Given  $x_0$ ; evaluate  $f_0, \nabla f_0$ ;  $p_0 \leftarrow -\nabla f_0$ ;  $k \leftarrow 0$ ;

**while**  $\nabla f_k \neq 0$ :

    compute  $\alpha_k$  by line search,  $x_{k+1} = x_k + \alpha_k p_k$ ;    evaluate  $\nabla f_{k+1}$ ;

$$\beta_{k+1}^{\text{FR}} \leftarrow \frac{\nabla f_{k+1}^T \nabla f_{k+1}}{\nabla f_k^T \nabla f_k}, \quad p_{k+1} \leftarrow -\nabla f_{k+1} + \beta_{k+1}^{\text{FR}} p_k, \quad k \leftarrow k + 1. \quad (5.41)$$

- ▶ On a strongly convex quadratic with exact line search, this *is* Algorithm 5.2.
- ▶ Per iteration:  $f$  and  $\nabla f$  only. No matrix, a few vectors. Built for large  $n$ .
- ▶ **Open question:** is  $p_{k+1}$  a descent direction? Not automatically.

# Descent needs a strong Wolfe line search with $c_2 < \frac{1}{2}$

Take the inner product of (5.41b) with  $\nabla f_k$ :

$$\nabla f_k^T p_k = -\|\nabla f_k\|^2 + \beta_k^{\text{FR}} \nabla f_k^T p_{k-1}. \quad (5.42)$$

Exact line search:  $\nabla f_k^T p_{k-1} = 0$ , so descent. Inexact: the second term may dominate and  $p_k$  may be an *ascent* direction.

## Strong Wolfe with a tighter $c_2$

$$f(x_k + \alpha_k p_k) \leq f_k + c_1 \alpha_k \nabla f_k^T p_k, \quad |\nabla f(x_k + \alpha_k p_k)^T p_k| \leq -c_2 \nabla f_k^T p_k, \quad (5.43)$$

with  $0 < c_1 < c_2 < \frac{1}{2}$  — not  $c_2 < 1$  as in Ch. 3.

## Lemma 5.6

Under (5.43) with  $c_2 < \frac{1}{2}$ , every FR direction is a descent direction, and

$$-\frac{1}{1 - c_2} \leq \frac{\nabla f_k^T p_k}{\|\nabla f_k\|^2} \leq \frac{2c_2 - 1}{1 - c_2} < 0. \quad (5.53)$$

Induction on  $k$  using (5.42), (5.43b), and the formula for  $\beta^{\text{FR}}$ . (Homework.) This is why Ch. 3 said  $c_2 = 0.1$  for CG: the typical 0.9 is not allowed here.

# Choices of $\beta_k$

All coincide on a strongly convex quadratic with exact line search (since then  $\nabla f_k^T \nabla f_{k+1} = 0$ ). On general  $f$  with inexact line search they behave very differently.

| Name           | $\beta_{k+1}$                                                                                | Notes                                            |
|----------------|----------------------------------------------------------------------------------------------|--------------------------------------------------|
| FR (5.41a)     | $\frac{\ \nabla f_{k+1}\ ^2}{\ \nabla f_k\ ^2}$                                              | global convergence (Thm 5.7); can stall          |
| PR (5.44)      | $\frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{\ \nabla f_k\ ^2}$                    | best in practice; can <i>cycle</i> (Thm 5.8)     |
| PR+ (5.45)     | $\max\{\beta_{k+1}^{\text{PR}}, 0\}$                                                         | PR with a built-in restart; convergent           |
| HS (5.46)      | $\frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{(\nabla f_{k+1} - \nabla f_k)^T p_k}$ | conjugacy w.r.t. the average Hessian $\bar{G}_k$ |
| FR-PR (5.48)   | $\beta^{\text{PR}}$ clipped to $[-\beta^{\text{FR}}, \beta^{\text{FR}}]$                     | PR behavior, FR guarantee                        |
| (5.49), (5.50) | Dai-Yuan; Hager-Zhang                                                                        | descent under plain Wolfe; competitive           |

HS derivation: require  $p_{k+1}^T \bar{G}_k p_k = 0$  where  $\bar{G}_k = \int_0^1 \nabla^2 f(x_k + \tau \alpha_k p_k) d\tau$ ; since  $\nabla f_{k+1} = \nabla f_k + \alpha_k \bar{G}_k p_k$  (Taylor, integral form), the denominator appears. Global convergence can be shown for any  $|\beta_k| \leq \beta_k^{\text{FR}}$  (5.47).

# Restarts

Theorem 5.3 needs  $p_0 = -r_0$ : the good properties hold only from a steepest-descent start. So **restart**: set  $\beta_k = 0$  periodically.

- ▶ Restart every  $n$  steps  $\Rightarrow$  *n-step quadratic convergence*:  $\|x_{k+n} - x^*\| = O(\|x_k - x^*\|^2)$  (5.51). Reason: near  $x^*$  the function is nearly quadratic; after a restart inside that region the method is nearly Algorithm 5.2, which finishes in  $n$  steps.
- ▶ In practice  $n$  is huge and an acceptable solution is found long before  $n$  steps — so restarts by count may never fire. Instead, restart on *loss of orthogonality*:

$$\frac{|\nabla f_k^T \nabla f_{k-1}|}{\|\nabla f_k\|^2} \geq \nu, \quad \nu \approx 0.1. \quad (5.52)$$

(On a quadratic, consecutive gradients are orthogonal; when they stop being so, the quadratic model has expired.)

- ▶ PR+ is an implicit restart: whenever  $\beta^{\text{PR}} < 0$  the direction reverts to  $-\nabla f$ . Rare, because  $\beta^{\text{PR}}$  is usually positive.
- ▶ Implementations also keep the quadratic/cubic interpolation of Ch. 3 in the line search, so that on a quadratic the exact  $\alpha_k$  is recovered and the method *is* linear CG.

# Why FR stalls and PR does not

From Lemma 5.6, with  $\cos \theta_k$  the angle to  $-\nabla f_k$ ,

$$\frac{1 - 2c_2}{1 - c_2} \frac{\|\nabla f_k\|}{\|p_k\|} \leq \cos \theta_k \leq \frac{1}{1 - c_2} \frac{\|\nabla f_k\|}{\|p_k\|}, \quad (5.57)$$

so  $\cos \theta_k \approx 0$  exactly when  $\|\nabla f_k\| \ll \|p_k\|$  — a bad, nearly orthogonal direction.

## FR

Bad direction  $\Rightarrow$  tiny step  $\Rightarrow \nabla f_{k+1} \approx \nabla f_k$

$\Rightarrow \beta_{k+1}^{\text{FR}} \approx 1 \Rightarrow p_{k+1} \approx p_k$ .

*The bad direction is inherited.* A long run of useless iterations follows.

## PR (and PR+, HS, FR-PR)

Bad direction  $\Rightarrow$  tiny step  $\Rightarrow \nabla f_{k+1} \approx \nabla f_k$

$\Rightarrow \beta_{k+1}^{\text{PR}} \approx 0 \Rightarrow p_{k+1} \approx -\nabla f_{k+1}$ .

*Automatic restart.*  $\cos \theta_{k+1} \approx 1$ .

Documented case,  $n = 100$ :  $\cos \theta_k \sim 10^{-2}$  and steps  $\sim 10^{-2}$  for hundreds of iterations; FR needs thousands of iterations, **PR needs 37**. Periodic restarts fix FR. *Never run FR without a restart strategy.*

# Global convergence (Theorems 5.7, 5.8)

Assumptions 5.1: level set bounded;  $f$  Lipschitz continuously differentiable near it. Zoutendijk (Ch. 3) then gives  $\sum \cos^2 \theta_k \|\nabla f_k\|^2 < \infty$  (5.60).

## Theorem 5.7 (Al-Baali)

FR with strong Wolfe steps,  $0 < c_1 < c_2 < \frac{1}{2}$ , satisfies  $\liminf_k \|\nabla f_k\| = 0$ .

Idea: if  $\|\nabla f_k\| \geq \gamma$  throughout, Lemma 5.6 lets one bound  $\|p_k\|^2 \leq c_3 \bar{\gamma}^4 k / \gamma^2$  (5.68); then (5.60) forces  $\sum 1/\|p_k\|^2 < \infty$ , but  $\sum 1/k = \infty$ . Contradiction. Extends to any  $|\beta_k| \leq \beta_k^{\text{FR}}$ , hence to FR-PR. Restarted methods: subsequence only.

## Theorem 5.8 (Powell) — the bizarre part

There is a  $C^2$  function on  $\mathbb{R}^3$  and an  $x_0$  such that PR with an *ideal* line search (first stationary point of  $\phi$ ) produces  $\|\nabla f_k\|$  bounded away from zero. PR can cycle.

The failure needs consecutive directions to become near-opposites, which under ideal line search requires  $\beta_k < 0$ . Hence **PR+**: set  $\beta = 0$  when negative. With a small modification of the Wolfe conditions, PR+ is globally convergent. Dai–Yuan (5.49) and Hager–Zhang (5.50) converge under plain Wolfe — no modification needed.

The best method in practice is the one for which the theory is weakest. This is normal.

## Numerical performance (Table 5.1, excerpt)

Strong Wolfe with  $c_1 = 10^{-4}$ ,  $c_2 = 0.1$ ; no restarts; stop at  $\|\nabla f_k\|_\infty < 10^{-5}(1 + |f_k|)$ ; failure (\*) after 10,000 iterations. Entries: iterations / function-gradient evaluations.

| Problem  | $n$  | FR        | PR        | PR+       | mod |
|----------|------|-----------|-----------|-----------|-----|
| CALCVAR3 | 200  | 2808/5617 | 2631/5263 | 2631/5263 | 0   |
| GENROS   | 500  | *         | 1068/2151 | 1067/2149 | 1   |
| XPOWSING | 1000 | 533/1102  | 212/473   | 97/229    | 3   |
| MSQRT1   | 1000 | 422/849   | 113/231   | 113/231   | 0   |
| TRIGON   | 1000 | 231/467   | 40/92     | 40/92     | 0   |

“mod” = number of iterations on which PR+ reset a negative  $\beta$  to zero.

- ▶ FR fails on GENROS: tiny steps far from the solution, no restart to break the cycle.
- ▶ PR/PR+ typically 2–5 $\times$  fewer evaluations; PR+ rarely differs from PR, and never for the worse here.
- ▶ Each iteration costs about two  $f/\nabla f$  evaluations — the line search is doing its job cheaply.

**Recommendation (the book’s):** PR, PR+, FR–PR, or the (5.49)/(5.50) methods. Not plain FR.

# Chapter 5 in one table

|              | Linear CG (§5.1)                                     | Nonlinear CG (§5.2)                         |
|--------------|------------------------------------------------------|---------------------------------------------|
| Problem      | $Ax = b, A \succ 0$                                  | $\min f$ , general                          |
| Direction    | $p_k = -r_k + \beta_k p_{k-1}$                       | $p_k = -\nabla f_k + \beta_k p_{k-1}$       |
| Step         | exact: $r_k^T r_k / p_k^T A p_k$                     | strong Wolfe, $c_2 < \frac{1}{2}$           |
| $\beta_k$    | $r_k^T r_k / r_{k-1}^T r_{k-1}$                      | FR / PR / PR+ / HS / ...                    |
| Key property | conjugate directions, Krylov optimality              | descent via Lemma 5.6                       |
| Termination  | $\leq n$ steps; $\leq r$ for $r$ eigenvalue clusters | $\liminf \ \nabla f_k\  = 0$ (FR, PR+)      |
| Rate         | $\sqrt{\kappa}$ instead of $\kappa$ ; clustering     | $n$ -step quadratic with restarts           |
| Cost / iter  | one $Ap$                                             | one $f$ , one $\nabla f$ (plus line search) |
| Storage      | 4 vectors                                            | a few vectors                               |
| Accelerator  | preconditioning $M \approx A$                        | restarts; choice of $\beta$                 |

## Three ideas to carry forward

*Conjugacy decouples an  $n$ -dimensional problem into  $n$  one-dimensional ones. Convergence depends on the eigenvalue distribution, not on  $n$  — so change the distribution. The Krylov optimality of Theorem 5.2 is why CG can be stopped early (Steihaug, Newton-CG).*

# Homework

## Derivations

- 1 Show that a conjugate set (w.r.t.  $A \succ 0$ ) is linearly independent.
- 2 Derive the  $\alpha_k, \beta_{k+1}$  of Algorithm 5.2 from Algorithm 5.1 using (5.10), (5.11), (5.16).
- 3 Prove Theorem 5.4 by constructing  $Q_r$  explicitly.
- 4 Show  $\beta^{\text{FR}} = \beta^{\text{PR}} = \beta^{\text{HS}}$  on a strongly convex quadratic with exact line search.
- 5 Prove Lemma 5.6 and derive the bounds (5.57) on  $\cos \theta_k$ .
- 6 Derive Algorithm 5.3 from Algorithm 5.2 on (5.38), eliminating  $C$  in favor of  $M = C^T C$ .

## Implementation

7. CG and PCG on the 2-D Poisson system ( $n \approx 10^4$ ),  $M = I$  vs. zero-fill incomplete Cholesky; plot  $\|r_k\|$ . Then build  $A$  with exactly 5 distinct eigenvalues and confirm Theorem 5.4.
8. Nonlinear CG (FR and PR+) with the Ch. 3 strong-Wolfe search,  $c_2 = 0.1$ , on extended Rosenbrock,  $n = 100$ . Log  $\cos \theta_k, \beta_k$ ; find where FR stalls and PR+ does not.
9. Return to Chapter 4, item 8: implement CG–Steihaug and rerun trust-region Newton.

Next: Ch. 6 — *quasi-Newton: learning the Hessian from gradients*.